



Celeste RL Agent

Bee Dahlgren

California Baptist University



Celeste

Precision Platformer



Everest

- Open-source mod loader and modding API
- Custom mods
- Map packs
- Texture replacement



Gym Bridge

Celeste Modding Experience

Movement

- Direct Physics Interactions
- TAS-Style Inputs
 - CelesteTAS-EverestInterop
- Accessing Hidden Methods
 - Reflection

GAMEPLAY

Left



Right



Up



Down



Jump



Dash



Grab



Talk



Observation

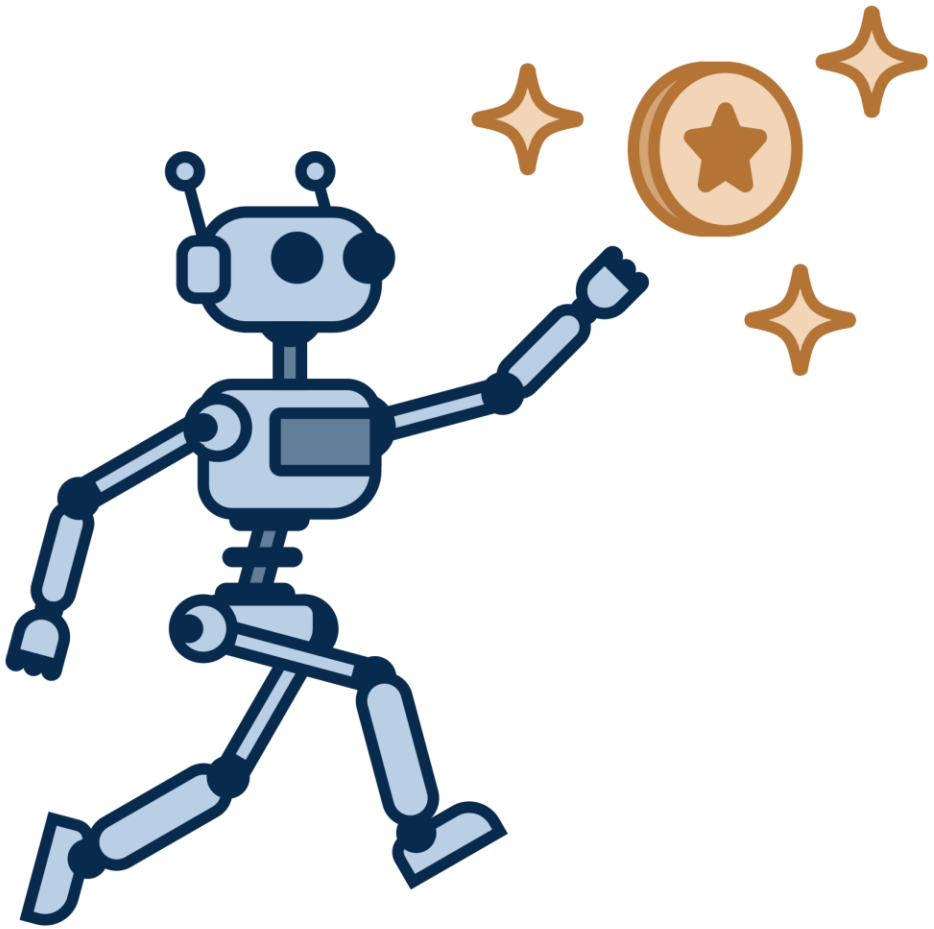
- Player Position
- Player Velocity
- Dashes
- Grab Toggled
- On Ground
- Facing
- Exit Position
- Vision Grid



RL Agent

Subtitle

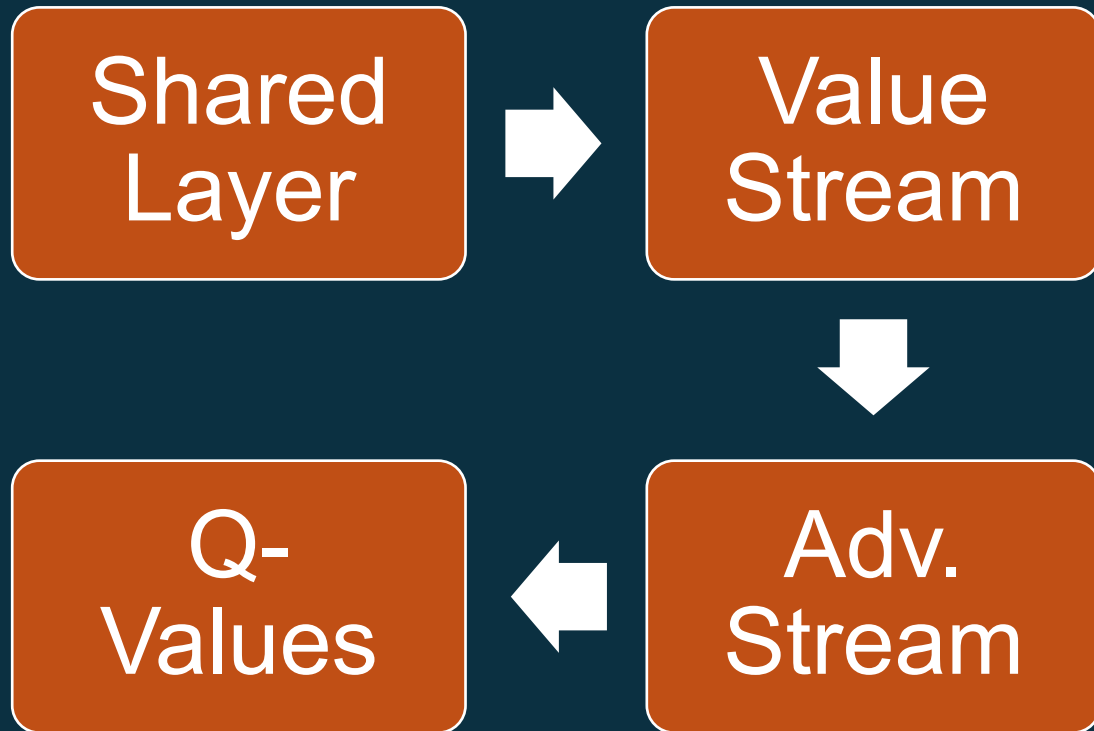
Celeste Env Server - Reward



- Positive
 - Movement Toward Exit
 - Survival
 - Big Rewards
 - Exit!
 - Checkpoint!
- Negative
 - Death
 - Stuck

Reinforcement Learning

Dueling DQN



PPO

DQN

Double DQN

Results

Outcomes

Successes

- Gym Bridge
- Test-level completion

Failures

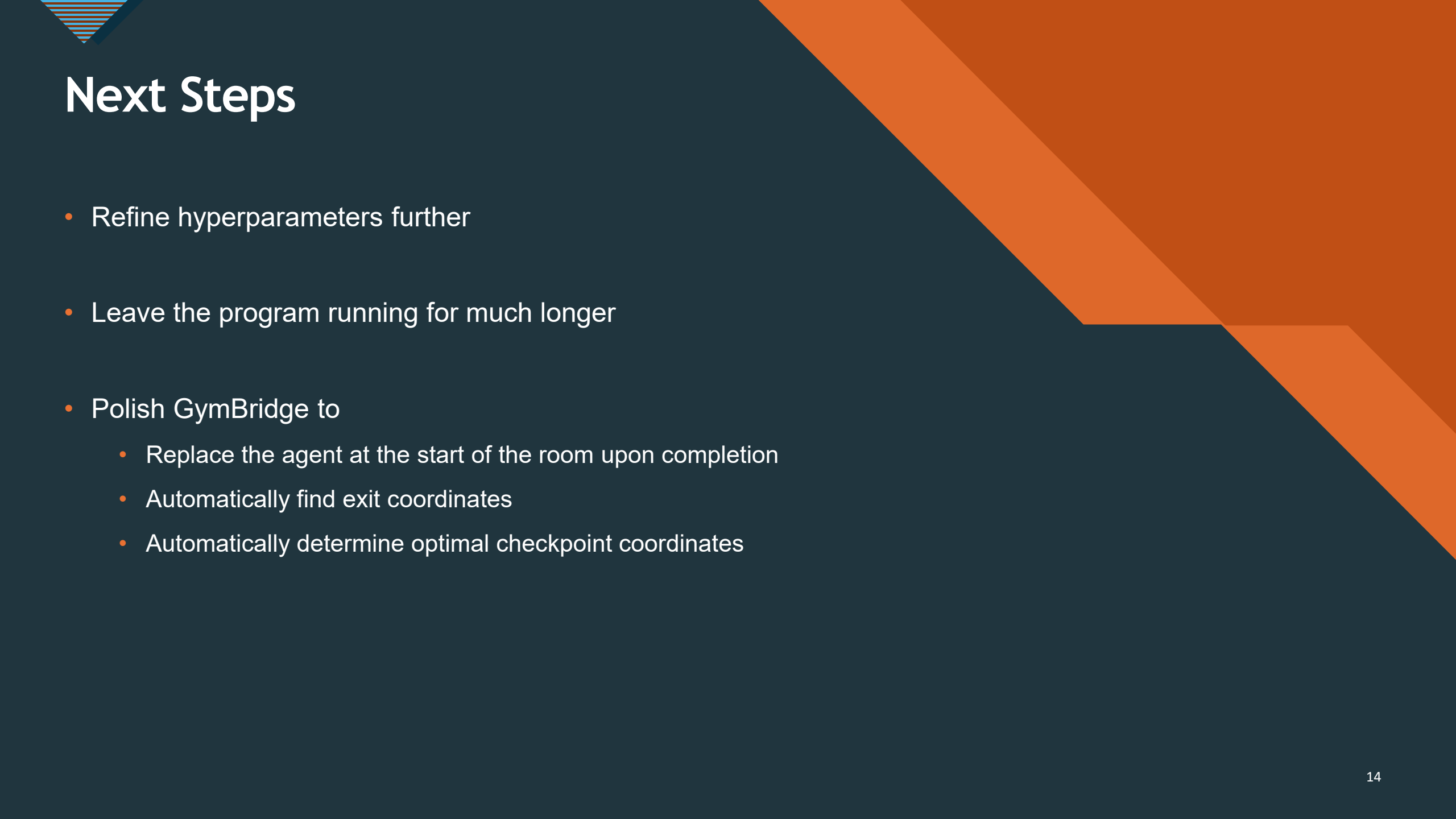
- Consistency
- Hard-coded exit

Pre-set Success Metrics

- Agent can complete a simple Celeste level at least 70% of the time.
 - Not met
- Demonstrated improvement in learning curve across training.
 - Met

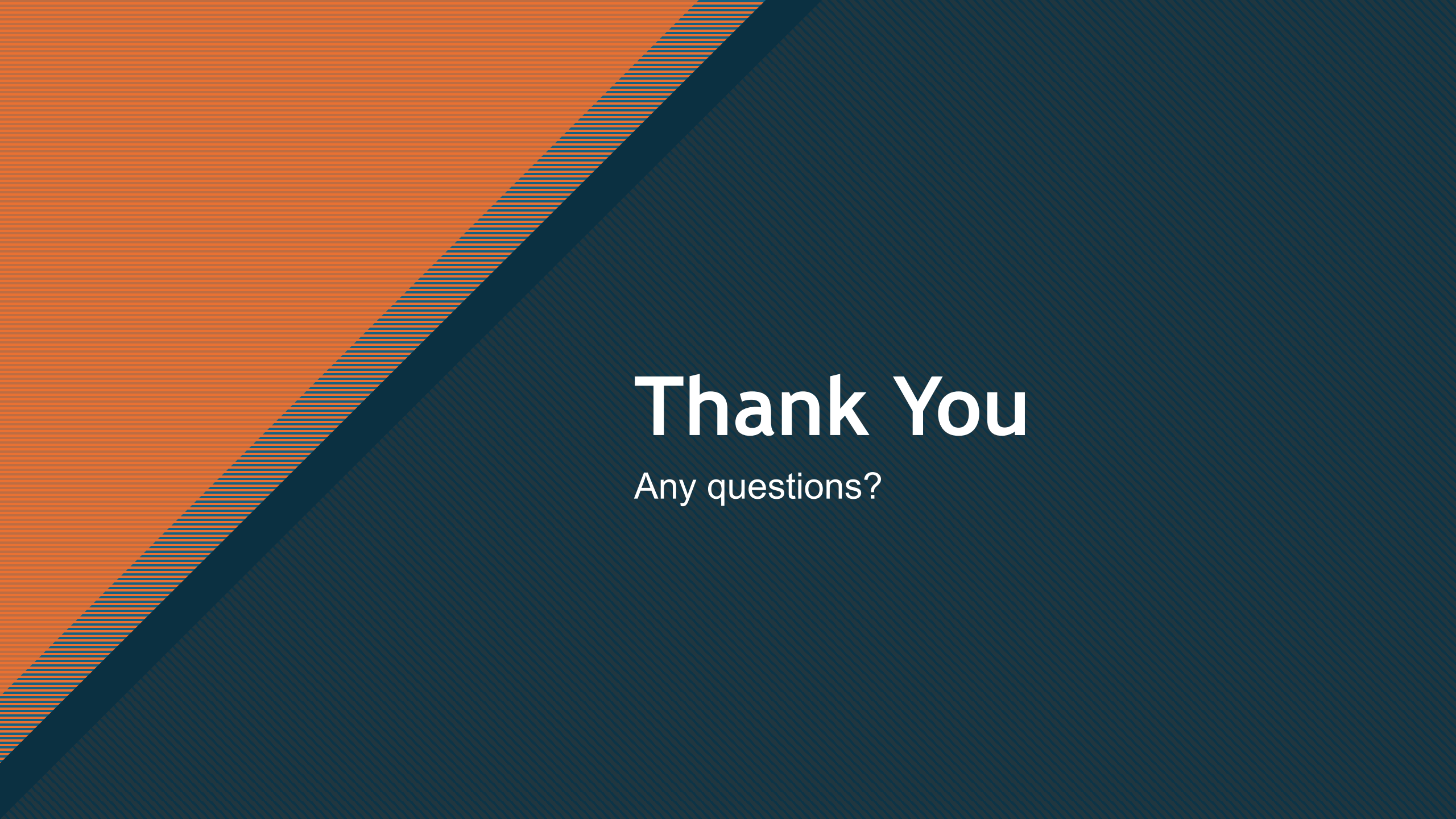
Live Demo

And recorded level completion



Next Steps

- Refine hyperparameters further
- Leave the program running for much longer
- Polish GymBridge to
 - Replace the agent at the start of the room upon completion
 - Automatically find exit coordinates
 - Automatically determine optimal checkpoint coordinates



Thank You

Any questions?